The Model Was Nearby, Not Involved

Revision-Exact Evidence Attribution for Asynchronous Industrial Intelligence

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Abstract

An industrial diagnostic interface may display a sensor observation beside a model prediction and thereby imply that the prediction was computed from that observation. In asynchronous edge systems, this inference is frequently invalid: acquisition, buffering, feature generation, model execution, caching, and case assembly advance on different schedules. We argue that proximity in wall-clock time is not sufficient evidence of computational attribution. We propose a Revision-Exact Attribution Contract (REAC), a compact provenance and decision protocol that classifies a displayed model result as exact-input, derived-input, temporally adjacent, or unrelated/indeterminate. The contract combines immutable acquisition identities, canonical input digests, execution attestations, clock-uncertainty intervals, and explicit abstention. We derive failure examples, compare provenance graphs, distributed tracing, and timestamp joins, and specify falsifiable replay and fault-injection tests. This is an architectural proposal, not an empirical demonstration of improved diagnostic accuracy.

Keywords: industrial edge AI; provenance; asynchronous inference; attribution; causal ordering; sensor lineage; evidence integrity; abstention; observability.

1. Problem statement and scope

A case investigator often juxtaposes three independently produced artifacts: a physical sensor measurement, a feature vector, and an inference output. A shared timestamp, machine label, or case identifier does not establish that the inference consumed that measurement. The distinction matters when operators interpret a model output as corroboration of a particular vibration burst, thermal event, or equipment transition. A misleading linkage can create false diagnostic confidence without any false value in the underlying telemetry.

Consider an acquisition stream S with observations s(101), s(102), and s(103). An inference worker consumes feature revision f(101), but a case viewer loads the newest observation s(103) and the newest cached inference y(101). The pair may differ by only 200 ms. Even perfect clock synchronization cannot turn y(101) into evidence about s(103). This is an attribution error, not necessarily a model error.

2. Thesis and epistemic boundary

Thesis. A system should assert that a model output pertains to a particular observation only when an auditable derivation path connects the output to the exact input revision, or when the claim is explicitly weakened to a bounded temporal association. No amount of timestamp proximity alone proves

derivation.

The contract concerns computational ancestry and evidence presentation. It does not prove physical causality, sensor calibration, model correctness, or that the measured machine was the true source of a mixed signal. These require independent experiments and physical references. Provenance can faithfully preserve a false measurement [1].

3. Prior work and competing approaches

W3C PROV-O defines entities, activities, agents, and relations such as used and wasGeneratedBy, providing a natural representation for observation-to-feature-to-inference ancestry [1]. OpenTelemetry traces and span links describe execution dependencies across asynchronous components, but their presence alone does not prove that a specific sensor sample was included in a model tensor [2,3]. Lamport's happens-before relation distinguishes ordering from concurrent or incomparable events; wall-clock timestamps are weaker evidence for causal execution order [4]. NIST AI RMF frames traceability and measurement as governance and evaluation concerns, not substitutes for validity testing [5].

The simplest competitor is a time-window join, which is cheap and useful for exploratory dashboards but unsafe as a derivation assertion. A shared case ID improves grouping but can still contain stale inference. Full distributed tracing improves operational diagnosis but can omit application-level input membership. Content-addressed provenance offers stronger reproducibility at a storage and integration cost. REAC is a narrow decision layer atop these existing techniques, not a replacement for them.

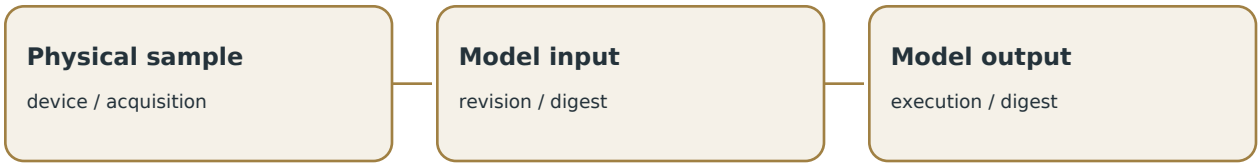
4. Formal attribution model

Let s=(device, acquisition_epoch, sample_counter, payload_digest) identify a physical acquisition; f=(pipeline_version, ordered_parent_ids, feature_digest) identify a derived input; and m=(model_digest, execution_id, input_digest, output_digest) identify an inference. Parent relationships are immutable edges in a directed acyclic provenance graph G. For a displayed observation s and output m, define $D(s,m)=1$ if there exists a verified directed path $s \rightarrow f \rightarrow m$ with exact matching input digest and admissible transformation declarations. This is computational derivation, not a claim that the sensor is physically accurate.

For two event-time estimates with uncertainty intervals $I(s)=[t_s-e_s,t_s+e_s]$ and $I(m)=[t_m-e_m,t_m+e_m]$, temporal overlap $I(s) \cap I(m) \neq \emptyset$ permits a proximity label only. It cannot establish D(s,m). Clock offset uncertainty, queue delays, and batch windows must be recorded separately; the inference timestamp usually marks execution, not acquisition.

A practical verdict function V(s,m) has four mutually exclusive outputs: EXACT_INPUT when verified ancestry and digest agreement exist; DERIVED_INPUT when a verified declared aggregation includes s but not as the sole input; PROXIMATE_ONLY when no verified path exists but the configured temporal relation holds; and UNRELATED_OR_UNKNOWN otherwise. A missing edge must never be silently upgraded into an exact-input assertion. An optional fifth flag, CONTRADICTORY, records positive evidence of a mismatched claimed revision.

5. Reference architecture



Exact lineage: signed/hashed edges between immutable acquisition, input, and execution IDs

Temporal proximity alone: dashed logical association, not an input-derivation claim

Acquisition emits a monotonic sample counter within an epoch; counter reuse after reboot requires a new epoch identity. Feature transforms record ordered parent IDs and versioned code/configuration digests. Inference execution records the exact tensor digest and model digest. The case investigator performs a graph join against immutable references, rather than pairing latest values. An append-only audit stream preserves attribution decisions and evidence used for each decision.

The proposed API response includes: observation_id, acquisition_epoch, sample_counter, source_id, feature_revision, feature_digest, model_id, model_digest, execution_id, input_digest, verdict, supporting_edges, clock_uncertainty_ms, and reason_codes. Hashes detect byte identity, not semantic equivalence; a canonical serialization specification is necessary to prevent equivalent tensors from producing different digests.

6. Failure analysis

Replay after reconnect. A collector republishes a cached sample with a new transport timestamp. Without epoch and sample-counter lineage, a time join may attribute an inference to a nonexistent new acquisition. Batch aggregation. A model consumes a window of 128 observations. A single sample may be a verified member, but presenting the result as uniquely caused by that sample is unwarranted. Concurrent workers. Two workers emit outputs in reverse completion order. Latest-result selection can attach the wrong execution to a case. Cross-source bleed. Exact computational ancestry can coexist with wrong physical source identification; REAC does not solve source separation. Digest collision or tampering. Cryptographic hashes require sound algorithms, trusted collection boundaries, and appropriate authentication.

7. Experimental protocol and falsification

We propose deterministic replay of 10,000 synthetic acquisition events across two sources, with randomized worker delay, queue reordering, reboot epochs, missing counters, batch windows, and controlled clock offsets. Ground-truth lineage is generated before transport perturbation. Compare (A) nearest-timestamp join, (B) case-ID join, (C) OpenTelemetry-only correlation, and (D) REAC. The primary metric is false exact-attribution rate, with confidence intervals; secondary metrics are abstention rate, precision/recall of exact links, end-to-end latency, metadata bytes per sample, storage growth, and investigator query time.

Hypothesis H1: under delayed or reordered processing, REAC yields fewer false exact attributions than timestamp joins. H2: REAC adds bounded overhead compatible with edge hardware under a specified sample rate. H3: abstention increases as lineage is deliberately removed. Reject H1 if a competing join

is equally accurate under matched information; reject H2 if overhead exceeds predeclared budgets; reject H3 if missing evidence is still labeled exact. No experimental result is claimed here.

8. Limitations and counterarguments

A complete provenance graph is expensive at high-rate vibration sampling; window-level lineage, Merkle aggregation, and bounded retention may be preferable, but weaken sample-level reconstruction. Legacy sensors may lack acquisition counters; the correct response is downgraded attribution, not synthetic certainty. Requiring exactness can reduce the number of seemingly useful case correlations, yet that reduction may represent honest uncertainty rather than lost information. Conversely, timestamp association is often entirely appropriate for exploratory trend analysis, provided its status is explicit. The proposal does not establish a general causal inference framework, and the taxonomy must be evaluated with actual operator decisions to show benefit.

9. Experimental scope and implementation boundary

This paper is a research proposal and evaluation protocol. Private operational checkpoints are excluded from this public edition. No implementation, deployment, field validation, or measured performance benefit is established by the proposal. A prototype should first be evaluated with recorded or simulated inputs, explicit baselines, failure cases, and independently measured outcomes.

10. Conclusion

A model output and an observation may be adjacent in a dashboard while remaining unrelated in computation. The smallest useful correction is not a larger confidence score but an explicit attribution verdict backed by revision-level evidence. A system that can say “nearby, not involved” is less rhetorically impressive and more scientifically defensible. Whether that discipline improves field diagnosis remains an empirical question.

Related SynapticSteel Technical Papers

STP-003, The Timestamp Is Not the Event (temporal semantics); STP-004, Silence Is a Signal, But Not a Measurement (sensor availability); STP-007, A Reconnection Is Not a Recovery (reacquisition); STP-009, The Nullspace Has No Name (physical source attribution). These are thematic relationships, not claims of external peer review.

References

[1] W3C. PROV-O: The PROV Ontology. W3C Recommendation, 2013. https://www.w3.org/TR/prov-o/

[2] OpenTelemetry. Tracing API specification, including span links. https://opentelemetry.io/docs/specs/otel/trace/api/

[3] OpenTelemetry. General semantic conventions for traces, metrics and events. https://opentelemetry.io/docs/specs/semconv/general/

[4] Lamport, L. Time, clocks, and the ordering of events in a distributed system. Communications of the ACM 21(7), 558–565, 1978. DOI:10.1145/359545.359563. https://doi.org/10.1145/359545.359563

[5] NIST. Artificial Intelligence Risk Management Framework, AI 100-1, 2023, and AI Resource Center. https://airc.nist.gov/

[6] W3C. PROV-DM: The PROV Data Model. W3C Recommendation, 2013. https://www.w3.org/TR/prov-dm/

[7] OpenTelemetry. Semantic Conventions overview. https://opentelemetry.io/docs/concepts/semantic-conventions/

Suggested citation and release control

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Author-released working paper. GitHub edition published 11 November 2026. Not peer reviewed; no empirical performance improvement or production deployment is established. No DOI is assigned to this GitHub edition. Cite its versioned repository record. Copyright Mark Yarian. All rights reserved; public reading and citation permitted.